

WINDOWS 11

HELP DESK TROUBLESHOOTING LAB

Hands-on IT Support Portfolio Project

VirtualBox Lab | Windows 11 | Tier 1 Help Desk / Service Desk

Project Summary

This portfolio documents hands-on Windows 11 troubleshooting performed in a controlled virtual lab. The focus is the troubleshooting process: identify the user's symptom, form a theory, gather evidence, apply a safe fix, verify the result, and document the outcome.

Technical Skills Demonstrated

- TCP/IP, DHCP, DNS, gateway and ping diagnostics
- Windows network adapter troubleshooting
- NTFS permissions and standard-user access
- Windows Services and startup state
- Task Manager and application recovery
- Storage, startup applications and Windows Update review
- Event Viewer analysis and event correlation
- Device Manager and audio-device troubleshooting
- Clear technical documentation and verification

Troubleshooting Method

Stage	What I Did
1. Identify	Clarified the symptom, impact, and what the user could or could not do.
2. Theory	Considered likely causes before changing settings.
3. Test	Used low-risk Windows tools and commands to gather evidence.
4. Fix	Applied the least disruptive corrective action appropriate to the lab.
5. Verify	Repeated the original test or had the user confirm functionality.
6. Document	Recorded findings, evidence, actions, and next steps.

Case 1: Network Connectivity - DNS Failure

Reported symptom: User reports that internet access is not working.

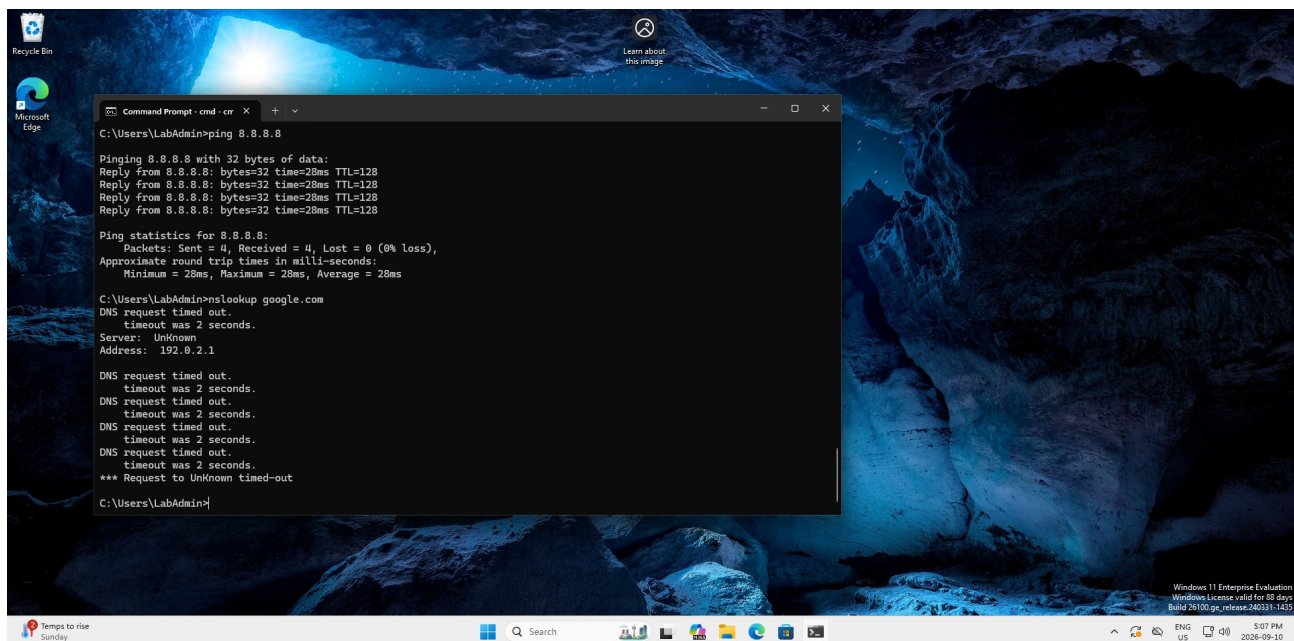
My troubleshooting reasoning: I would first confirm the physical connection and then use Command Prompt to test connectivity. I would compare the local IP configuration, default gateway, an external IP address, and a hostname so I can separate a general network failure from a DNS-specific failure.

Tools used: Command Prompt, ipconfig, ping

Findings: The diagnostic sequence showed that network connectivity could be tested independently from name resolution. This allowed the problem to be narrowed to DNS rather than treating every internet complaint as the same issue.

Corrective action: Corrected the DNS/network configuration used in the lab and repeated the connectivity tests.

Verification / next step: Successful responses after the correction confirmed restored name resolution and connectivity.



Evidence 1.1 - Network Connectivity - DNS Failure

```
Command Prompt - cmd - crr X + v
DHCPv6 Client DUID. . . . . : 00-01-00-01-32-2B-7D-E7-08-00-27-A6-BF-33
DNS Servers . . . . . : 64.59.184.13
                        64.59.190.242
                        fd17:625c:f037:2::3
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\LabAdmin>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=28ms TTL=128
Reply from 8.8.8.8: bytes=32 time=28ms TTL=128
Reply from 8.8.8.8: bytes=32 time=28ms TTL=128
Reply from 8.8.8.8: bytes=32 time=29ms TTL=128

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 29ms, Average = 28ms

C:\Users\LabAdmin>nslookup google.com
Server:  nsc1.ar.ed.shawcable.net
Address:  64.59.184.13

Non-authoritative answer:
Name:     google.com
Addresses: 2607:f8b0:400a:808::200e
          142.250.69.174

C:\Users\LabAdmin>
```

Evidence 1.2 - Network Connectivity - DNS Failure

Case 2: Network Connectivity - Disabled/Disconnected Adapter

Reported symptom: User reports no internet connection.

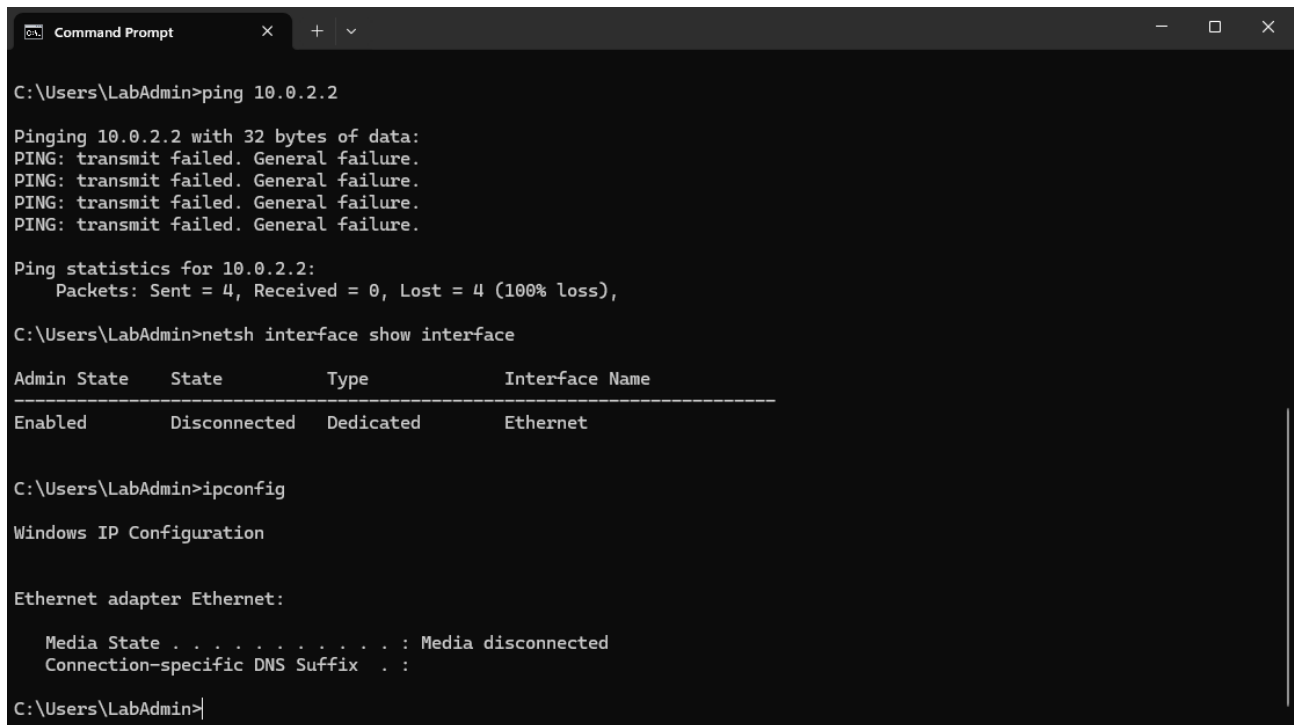
My troubleshooting reasoning: Before changing software settings, I would check the physical connection: cable, router connection, and link lights. If those look normal, I would check Windows network status and test from Command Prompt. When no usable Ethernet configuration appears and pings fail, I would investigate the network adapter state.

Tools used: Settings, Device Manager / Network Connections, Command Prompt, ipconfig, ping

Findings: The adapter was disconnected/disabled. The failed ping tests and missing usable interface configuration were consistent with the adapter not being active.

Corrective action: Enabled/restored the Ethernet adapter.

Verification / next step: A valid Ethernet configuration returned and pings to the gateway/external destination succeeded.



```
C:\Users\LabAdmin>ping 10.0.2.2

Pinging 10.0.2.2 with 32 bytes of data:
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.

Ping statistics for 10.0.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\LabAdmin>netsh interface show interface

Admin State   State          Type           Interface Name
-----
Enabled       Disconnected   Dedicated      Ethernet

C:\Users\LabAdmin>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\Users\LabAdmin>
```

Evidence 2.1 - Network Connectivity - Disabled/Disconnected Adapter

```
Command Prompt
C:\Users\LabAdmin>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : fd17:625c:f037:2:2b52:1c51:c363:2f88
    Temporary IPv6 Address. . . . . : fd17:625c:f037:2:890a:b45a:4f3d:8b14
    Link-local IPv6 Address . . . . . : fe80::92f4:a93f:396c:baa1%5
    IPv4 Address. . . . . : 10.0.2.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2%5
                                10.0.2.2

C:\Users\LabAdmin>ping 10.0.2.2

Pinging 10.0.2.2 with 32 bytes of data:
Reply from 10.0.2.2: bytes=32 time<1ms TTL=128
Reply from 10.0.2.2: bytes=32 time<1ms TTL=128
Reply from 10.0.2.2: bytes=32 time<1ms TTL=128
Reply from 10.0.2.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\LabAdmin>
```

Evidence 2.2 - Network Connectivity - Disabled/Disconnected Adapter

Case 3: Network Configuration - Incorrect Static IP

Reported symptom: The workstation has network problems even though the adapter itself is present.

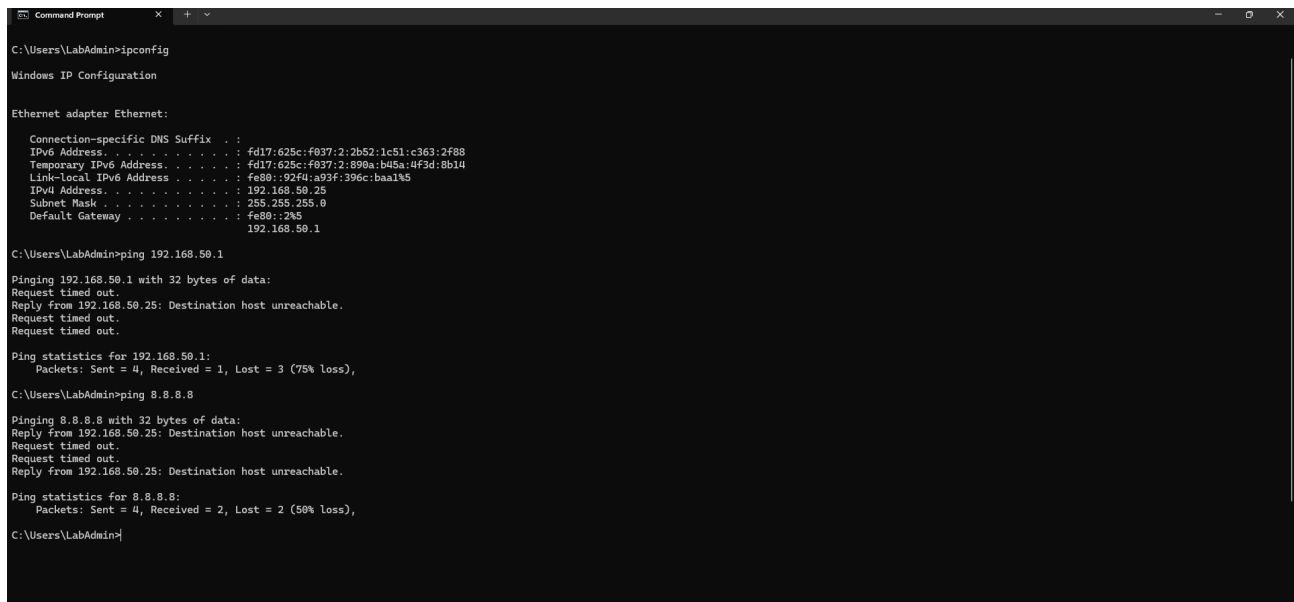
My troubleshooting reasoning: Once the adapter is active, I would inspect the assigned IP address, subnet mask, gateway, and DHCP state. A wrong static configuration can leave the adapter enabled while still preventing normal communication.

Tools used: Command Prompt, ipconfig, Windows network adapter settings

Findings: The lab workstation had an incorrect static IP configuration.

Corrective action: Returned the adapter to DHCP so it could obtain the correct network configuration automatically.

Verification / next step: A valid DHCP configuration was obtained and connectivity tests passed.



```
C:\Users\LabAdmin>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : fd17:625c:f037:2:2b52:1c51:c363:2f88
    Temporary IPv6 Address. . . . . : fd17:625c:f037:2:890a:b45a:4f3d:8b14
    Link-local IPv6 Address . . . . . : fe80::92f4:a99f:39ec:baa1%
    IPv4 Address. . . . . : 192.168.50.25
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2%5
                                192.168.50.1

C:\Users\LabAdmin>ping 192.168.50.1

Pinging 192.168.50.1 with 32 bytes of data:
Request timed out.
Reply from 192.168.50.25: Destination host unreachable.
Request timed out.
Request timed out.

Ping statistics for 192.168.50.1:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),

C:\Users\LabAdmin>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 192.168.50.25: Destination host unreachable.
Request timed out.
Request timed out.
Reply from 192.168.50.25: Destination host unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

C:\Users\LabAdmin>
```

Evidence 3.1 - Network Configuration - Incorrect Static IP

```
10.0.2.2

C:\Users\LabAdmin>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : fd17:625c:f037:2:2b52:1c51:c363:2f88
    Temporary IPv6 Address. . . . . : fd17:625c:f037:2:890a:b45a:4f3d:8b14
    Link-local IPv6 Address . . . . . : fe80::92f4:a93f:396c:baa1%5
    IPv4 Address. . . . . : 10.0.2.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2%5
                                10.0.2.2

C:\Users\LabAdmin>ping 10.0.2.2

Pinging 10.0.2.2 with 32 bytes of data:
Reply from 10.0.2.2: bytes=32 time<1ms TTL=128
Reply from 10.0.2.2: bytes=32 time<1ms TTL=128
Reply from 10.0.2.2: bytes=32 time<1ms TTL=128
Reply from 10.0.2.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\LabAdmin>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=28ms TTL=128
Reply from 8.8.8.8: bytes=32 time=28ms TTL=128
Reply from 8.8.8.8: bytes=32 time=28ms TTL=128
Reply from 8.8.8.8: bytes=32 time=28ms TTL=128

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 28ms, Average = 28ms

C:\Users\LabAdmin>
```

Evidence 3.2 - Network Configuration - Incorrect Static IP

Case 4: NTFS File Permissions - Employee Access

Reported symptom: A standard employee account cannot open a training document in C:\CompanyData\EmployeeFiles.

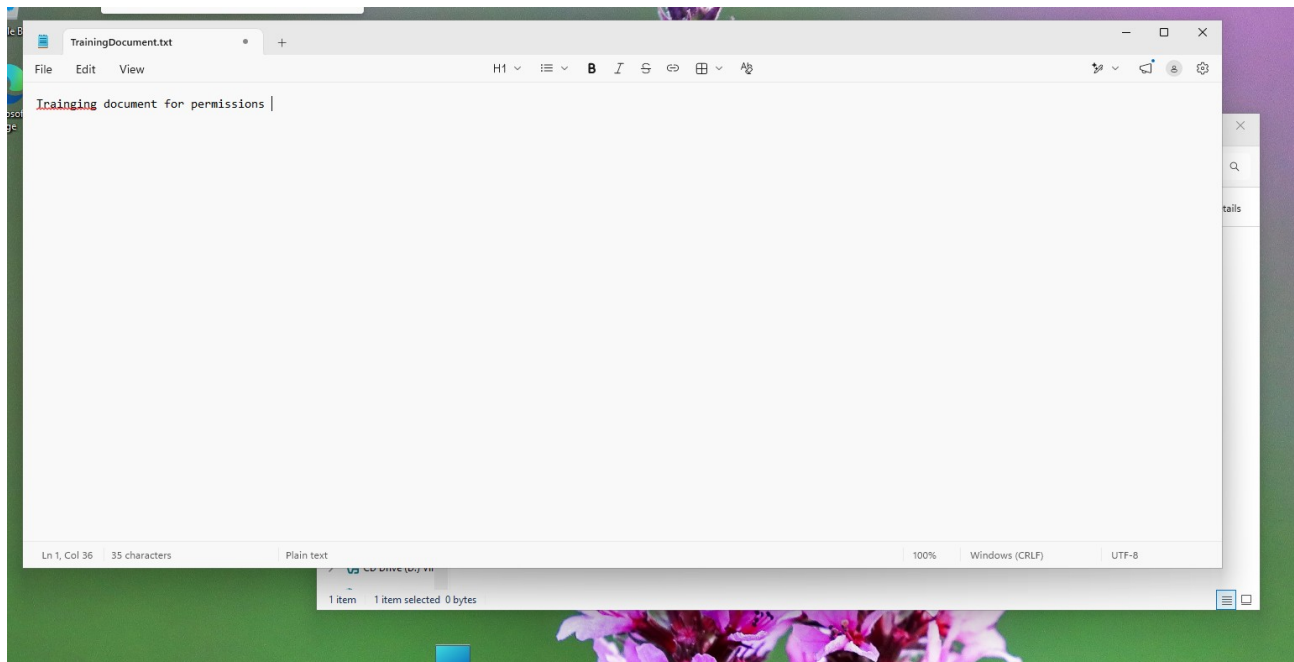
My troubleshooting reasoning: The access-denied message points toward permissions rather than a missing file. I would inspect the file/folder Security settings, inheritance, and the employee's effective permissions before changing ownership or granting broader access.

Tools used: File Explorer, Properties, Security, Advanced Security Settings

Findings: The standard user did not have the required access to the training document.

Corrective action: Adjusted the lab permissions to provide the required read access while avoiding unnecessary administrative/full-control rights.

Verification / next step: The TestEmployee account could open and read the training document after the permissions change.



Evidence 4.1 - NTFS File Permissions - Employee Access

Case 5: Windows Search Service - Search Not Working

Reported symptom: User reports that Windows Search is not functioning.

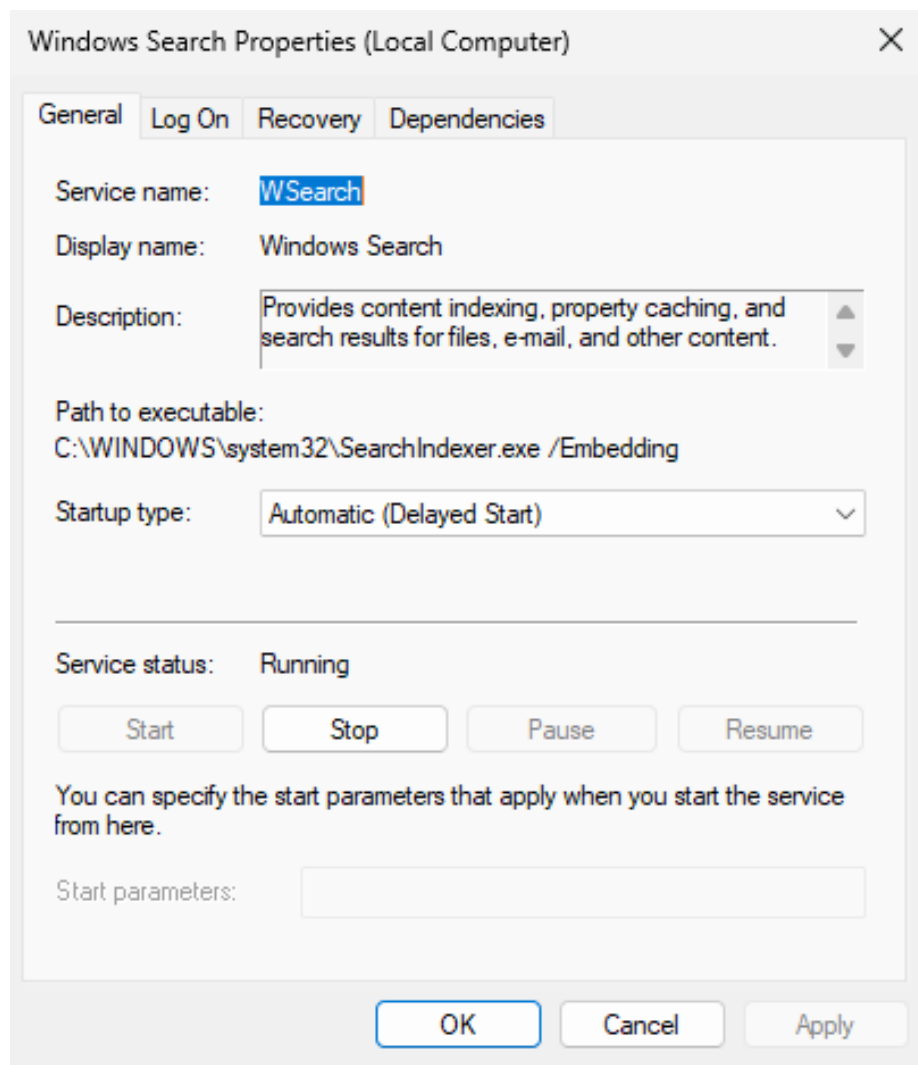
My troubleshooting reasoning: I would inspect the Windows Search service in services.msc. Service Status tells me whether it is currently running, while Startup Type determines how Windows starts it. If the service is stopped or disabled, that can explain the search problem.

Tools used: Services (services.msc)

Findings: Windows Search had been stopped/disabled in the lab scenario.

Corrective action: Restored the Windows Search service to an appropriate startup state and started the service.

Verification / next step: The service showed Running and Windows Search functionality was restored.



Evidence 5.1 - Windows Search Service - Search Not Working

Case 6: Application Troubleshooting - Frozen Notepad

Reported symptom: A user application becomes unresponsive.

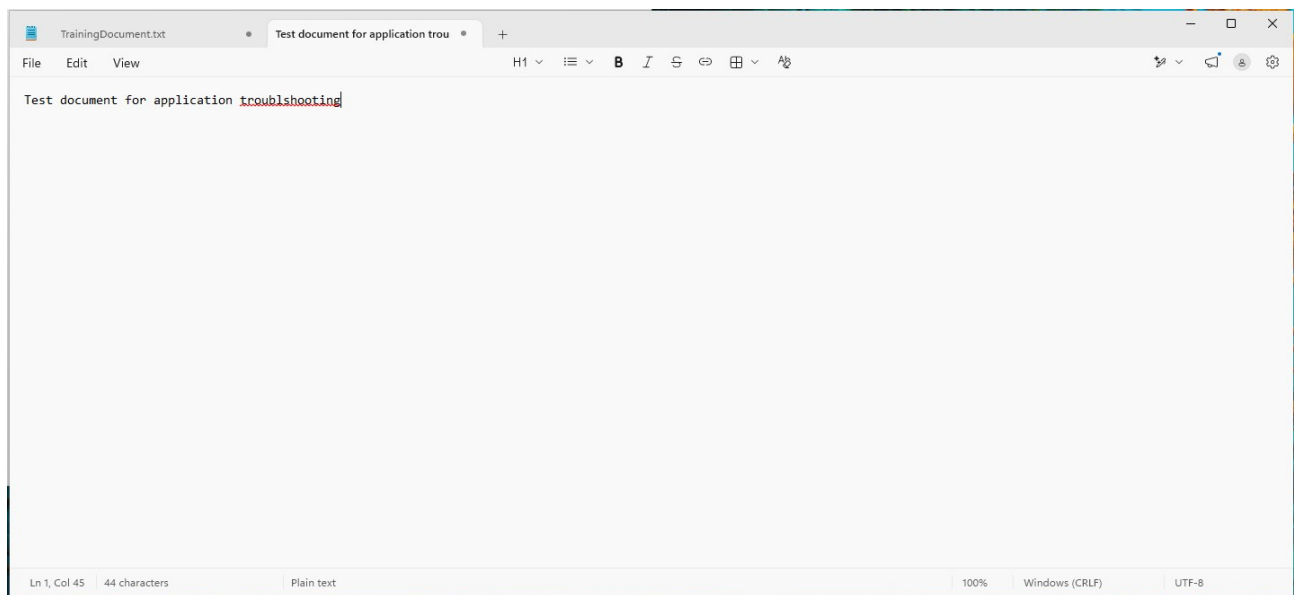
My troubleshooting reasoning: If only one application is frozen, I would avoid restarting the entire computer. I would open Task Manager, identify the affected process, and end only that application if it is safe to do so.

Tools used: Task Manager, Notepad

Findings: The issue was isolated to the application rather than the whole Windows session.

Corrective action: Force-closed the nonresponsive Notepad process and reopened the application.

Verification / next step: Notepad reopened and accepted input normally.



Evidence 6.1 - Application Troubleshooting - Frozen Notepad

Case 7: Performance and Stability Investigation

Reported symptom: User reports that the computer has recently been running slowly or has experienced an unexpected restart.

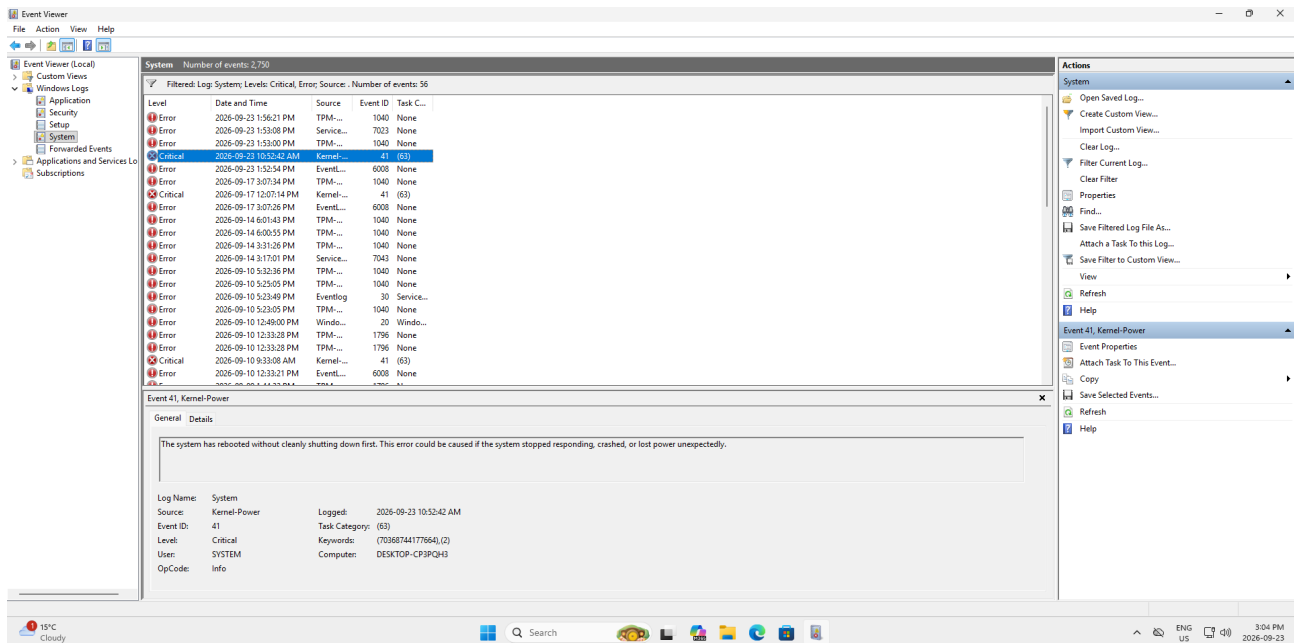
My troubleshooting reasoning: I would avoid assuming storage is the cause. With roughly 50 GB of free space, I would continue checking Task Manager resource usage, startup applications, Windows Update, and Event Viewer. A pending update may be relevant, but I would verify whether it is required before installing it. Event logs help establish whether a shutdown or crash occurred.

Tools used: Settings > Storage, Task Manager, Startup Apps, Windows Update, Event Viewer

Findings: No abnormal resource-heavy process was evident in the test VM. Event Viewer recorded Kernel-Power Event ID 41 and EventLog Event ID 6008, both consistent with an unexpected/unclean shutdown. These events document the shutdown but do not by themselves prove the underlying cause.

Corrective action: Documented the event IDs and correlated the timestamps. In a real ticket, I would continue with update history, hardware/power checks, reliability history, and recurrence data before assigning root cause.

Verification / next step: The VM was operational after restart; the investigation produced documented evidence and an escalation path rather than an unsupported conclusion.



Evidence 7.1 - Performance and Stability Investigation

Event Viewer (Local)

File Action View Help

Custom Views Windows Logs Application Security Setup System Forwarded Events Applications and Services Log Subscriptions

System Number of events: 2,750

Filtered Log: System; Levels: Critical, Error; Source: ... Number of events: 56

Level	Date and Time	Source	Event ID	Task C...
Error	2026-09-23 1:56:21 PM	TPM...	1040	None
Error	2026-09-23 1:53:08 PM	Service...	7023	None
Error	2026-09-23 1:53:00 PM	TPM...	1040	None
Critical	2026-09-23 10:53:42 AM	Kernel...	41 (63)	
Error	2026-09-23 1:52:54 PM	EventL...	6008	None
Error	2026-09-17 3:07:34 PM	TPM...	1040	None
Critical	2026-09-17 12:07:14 PM	Kernel...	41 (63)	
Error	2026-09-17 3:07:26 PM	EventL...	6008	None
Error	2026-09-14 6:01:43 PM	TPM...	1040	None
Error	2026-09-14 6:00:55 PM	TPM...	1040	None
Error	2026-09-14 3:31:26 PM	TPM...	1040	None
Error	2026-09-14 3:17:01 PM	Service...	7043	None
Error	2026-09-10 5:22:36 PM	TPM...	1040	None
Error	2026-09-10 5:25:05 PM	TPM...	1040	None
Error	2026-09-10 5:23:49 PM	Eventlog	30	Service...
Error	2026-09-10 5:23:05 PM	TPM...	1040	None
Error	2026-09-10 12:49:00 PM	Windo...	20	Windo...
Error	2026-09-10 12:33:28 PM	TPM...	1796	None
Error	2026-09-10 12:33:28 PM	TPM...	1796	None
Critical	2026-09-10 9:33:08 AM	Kernel...	41 (63)	
Error	2026-09-10 12:33:21 PM	EventL...	6008	None

Event 6008, EventLog

General Details

The previous system shutdown at 12:01:25 AM on 9/21/2026 was unexpected.

Log Name: System
Source: EventLog
Event ID: 6008
Level: Error
User: N/A
OpCode: Info

Logged: 2026-09-23 1:52:54 PM
Task Category: None
Keywords: Classic
Computer: DESKTOP-CP3PQH3

DESKTOP-CP3PQH3

15°C Cloudy

Search

ENG US 3:07 PM 2026-09-23

Evidence 7.2 - Performance and Stability Investigation

Case 8: Audio Troubleshooting - Disabled Output Device

Reported symptom: User reports no sound and Windows shows no output device.

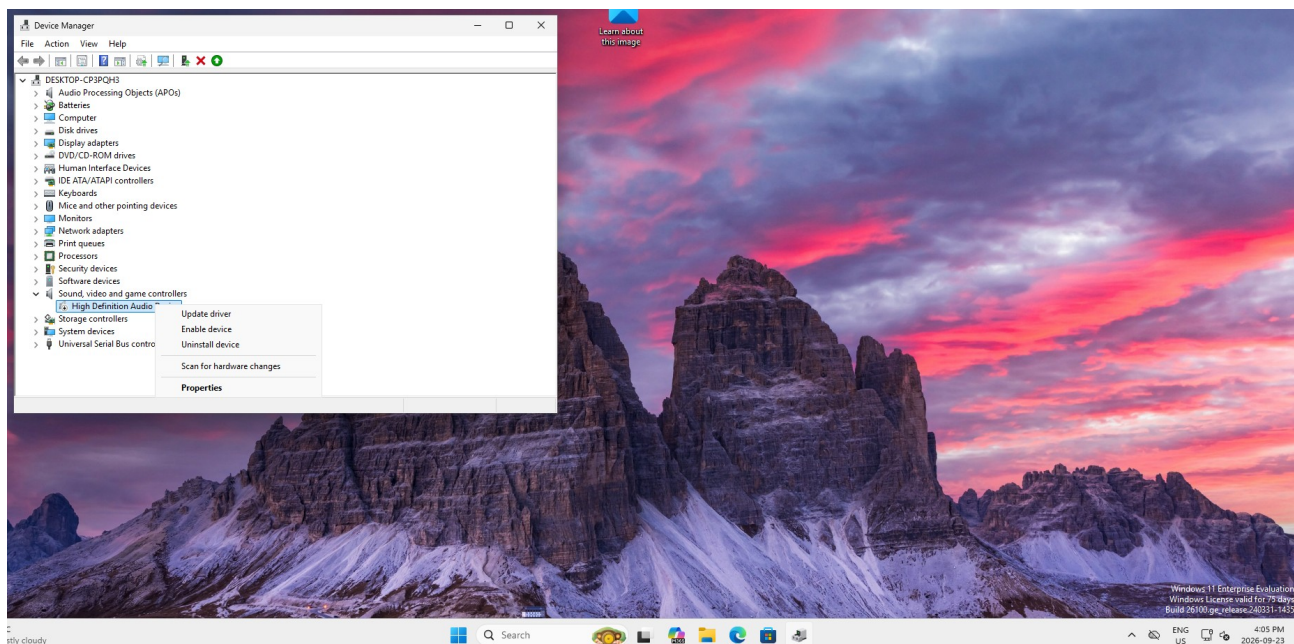
My troubleshooting reasoning: I would first confirm the physical setup: where the speakers are connected, whether cables are secure, and whether the expected output is the monitor or another device. If the physical setup is correct, I would check Sound settings and then Device Manager.

Tools used: Settings > System > Sound, Device Manager

Findings: The High Definition Audio Device was present but disabled. The context menu displayed 'Enable device,' confirming the device was not currently enabled.

Corrective action: Enabled the High Definition Audio Device.

Verification / next step: The next step is to have the user play audio and confirm sound output. The lab diagnosis and corrective action were completed.



Evidence 8.1 - Audio Troubleshooting - Disabled Output Device

What This Lab Demonstrates to an Employer

The strongest outcome of this lab is the troubleshooting discipline behind the fixes. I practiced separating symptoms from root causes, using evidence before making changes, avoiding unnecessary disruption, and verifying the user's original problem after the corrective action.

- I can explain why I chose a diagnostic step instead of only listing commands.
- I can distinguish network-layer problems such as adapter state, IP configuration, gateway reachability and DNS.
- I understand the difference between service status and startup configuration.
- I use least-privilege thinking when correcting file permissions.
- I can isolate a frozen application instead of immediately rebooting an entire workstation.
- I treat Event Viewer entries as evidence to correlate, not automatic proof of root cause.
- I verify the result from the user's perspective before closing a ticket.

Interview-Ready Summary

In my Windows 11 support lab, I worked through realistic Tier 1 incidents involving DNS, DHCP, network adapters, NTFS permissions, Windows services, frozen applications, unexpected shutdown events, and audio devices. For each scenario, I started with the user's symptom, gathered evidence using built-in Windows tools, made a targeted change, and verified the result. The lab helped me practice not just the technical fix, but the reasoning and documentation expected in a Help Desk environment.